

CHAPTER – 2

2. ENTITY ANALYSIS

2.1 Introduction

This section identifies the entities, attributes, primary keys, foreign keys, and relationships required for the **Titan Eye Eyewear E-Commerce System**. These database components provide the foundation for designing an efficient and well-structured Database Management System (DBMS). The system manages customer information, eyewear products, orders, payments, inventory, and deliveries.

2.2 Entity Identification

The following entities have been identified for the Titan Eye Eyewear E-Commerce System:

- Customer
- Product
- Category
- Brand
- Order
- Order Item
- Payment
- Shipment
- Admin

2.3 Entity Attributes

2.3.1 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Contact number
Address	Delivery address
Password	Customer login password

2.3.2 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Eyewear product name
Price	Product price
Stock	Available stock
Category_ID	Category ID
Brand_ID	Brand ID
Frame_Color	Frame color
Lens_Type	Lens type

2.3.3 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name (Sunglasses, Eyeglasses, Contact Lenses, etc.)

2.3.4 BRAND TABLE

Attribute	Description
Brand_ID	Unique brand ID
Brand_Name	Brand name

2.3.5 ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

2.3.6 ORDER ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2.3.7 PAYMENT TABLE

Attribute	Description
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Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	UPI/Card/Cash on Delivery
Payment_Status	Paid/Pending

2.3.8 SHIPMENT TABLE

Attribute	Description
Shipment_ID	Unique shipment ID
Order_ID	Order ID
Tracking_Number	Tracking number
Delivery_Status	Delivered/In Transit/Pending

2.3.9 ADMIN TABLE

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address
Password	Login password

2.4 Primary Keys

Entity	Primary Key
Customer	Customer_ID

Product	Product_ID
Category	Category_ID
Brand	Brand_ID
Order	Order_ID
Order Item	Order_Item_ID
Payment	Payment_ID
Shipment	Shipment_ID
Admin	Admin_ID

2.5 Foreign Keys

Table	Foreign Key	References
Product	Category_ID	Category
Product	Brand_ID	Brand
Order	Customer_ID	Customer
Order Item	Order_ID	Order
Order Item	Product_ID	Product
Payment	Order_ID	Order
Shipment	Order_ID	Order

2.6 Relationships

Parent Entity	Relationship	Child Entity
Customer	One-to-Many	Order

Category	One-to-Many	Product
Brand	One-to-Many	Product
Order	One-to-Many	Order Item
Product	One-to-Many	Order Item
Order	One-to-One	Payment
Order	One-to-One	Shipment

2.6.1 Relationship Explanation

Relationship	Cardinality	Explanation
Customer → Order	One-to-Many (1:M)	One customer can place multiple orders.
Category → Product	One-to-Many (1:M)	One category contains multiple eyewear products.
Brand → Product	One-to-Many (1:M)	One brand offers multiple eyewear products.
Order → Order Item	One-to-Many (1:M)	One order can contain multiple products.
Product → Order Item	One-to-Many (1:M)	One product can appear in multiple orders.
Order → Payment	One-to-One (1:1)	Each order has one payment record.
Order → Shipment	One-to-One (1:1)	Each order has one shipment record.

2.7 Conclusion

The identified entities, attributes, primary keys, foreign keys, and relationships provide the foundation for designing the Titan Eye Eyewear E-Commerce Database System. These components ensure efficient management of customers, eyewear products, brands, orders, payments, and deliveries, and will be used to create the Entity-Relationship (ER) Diagram and relational database design in the next phase of the project.

This version is suitable for a college mini project and closely follows the structure of your uploaded document while being customized for Titan Eye Eyewear.

